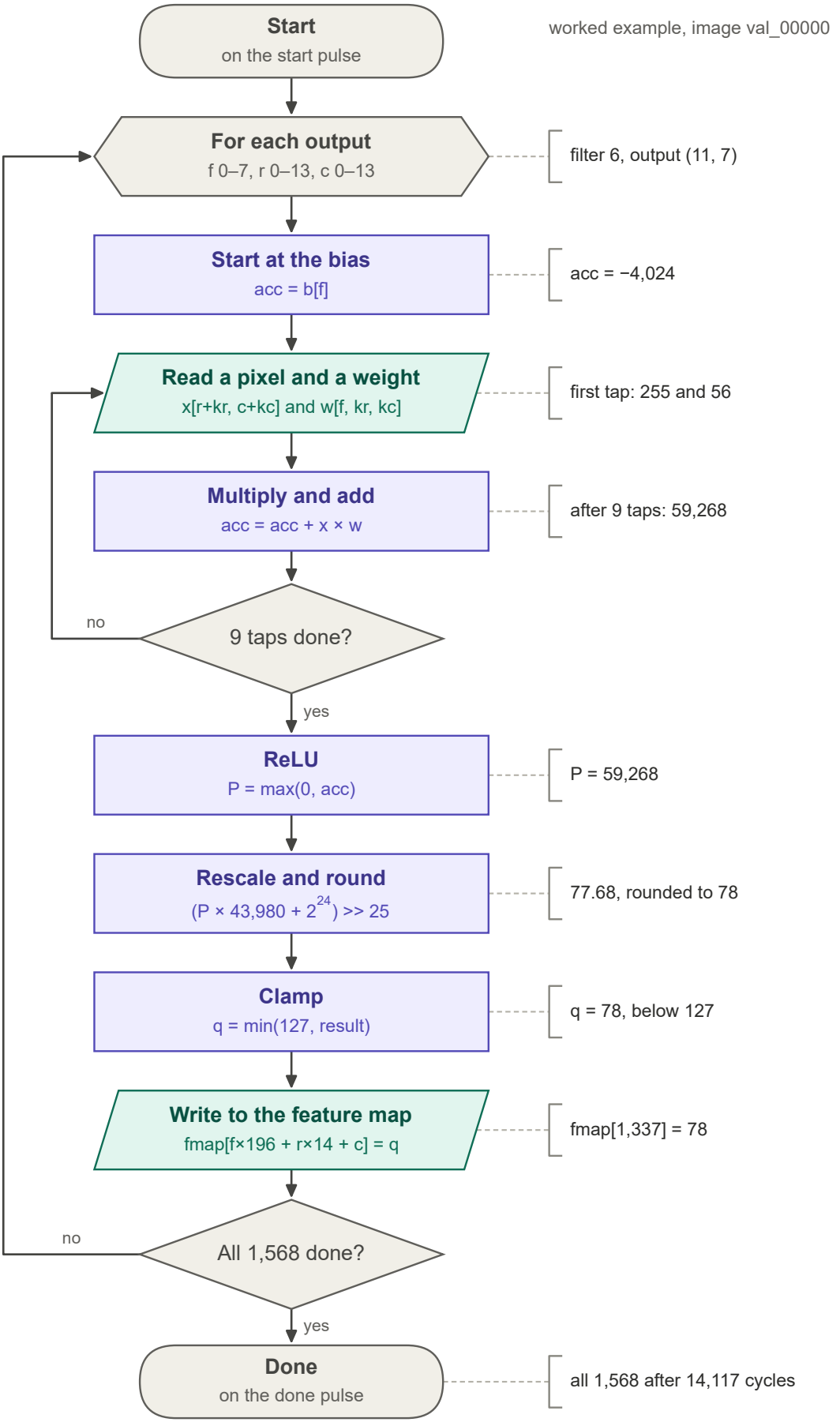
worked example, image val_00000
Start
on the start pulse
For each output
f 0–7, r 0–13, c 0–13
filter 6, output (11, 7)
Start at the bias
acc = b[f]
acc = −4,024
Read a pixel and a weight
x[r+kr, c+kc] and w[f, kr, kc]
first tap: 255 and 56
Multiply and add
acc = acc + x × w
after 9 taps: 59,268
no
9 taps done?
yes
ReLU
P = max(0, acc)
P = 59,268
Rescale and round
(P × 43,980 + 2^24) >> 25
77.68, rounded to 78
Clamp
q = min(127, result)
q = 78, below 127
Write to the feature map
fmap[f×196 + r×14 + c] = q
fmap[1,337] = 78
no
All 1,568 done?
yes
Done
on the done pulse
all 1,568 after 14,117 cycles